Chapter # 7 – Expert Systems.

June 2017- P11

2 Tick the **four** statements which are true about the use of expert systems.

	✓
A knowledge base is a key component of an expert system.	
Experts do not need to make a judgement; they just accept the results of the expert system.	
The user interface is used for both input and output.	
The rules base can contain IF… THEN statements.	
The inference engine searches the user interface to find possible solutions.	
An expert system is often used in car driving simulators.	
An expert system is not as accurate as a single expert would be.	
A knowledge engineer is employed to create an expert system.	
An expert system never needs updating.	
Expert systems always come up with exact solutions.	

[4]

Question 2

Answer:

2	A knowledge base is a key component of an expert system.	✓	**4**
	Experts do not need to make a judgement. They just accept the results of the expert system.		
	The user interface is used for both input and output.	✓	
	The rules base can contain IF…THEN statements.	✓	
	The inference engine searches the user interface to find possible solutions.		
	An expert system is often used in car driving simulators.		
	An expert system is not as accurate as a single expert would be.		
	A knowledge engineer is employed to create an expert system.	✓	
	An expert system never needs updating.		
	Expert systems always come up with exact solutions.		

November 2017 – P11 & P12

11 A space agency pays scientists whose job is to enable rockets to be sent to the moon.
Describe how batch and real-time processing would be used by the agency and its scientists.

Answer:

11	***Eight** from:* Batch processing would be used by payroll department to pay wages Batch processing would be used if the scientists had collected a very large amount of data offline and need to now process it all in one go Transaction file of hours worked is kept Master file of workers details/rate of pay per hour Transaction file is used with master file to update master file/produce payslips Jobs are set up so they can be run to completion without human interaction The input data are collected into batches and each batch is processed as a whole Batch processing can occur when the computing resources are less busy Batches can be stored up during working hours and then executed during the evening/whenever the computer is idle Batch processing is particularly useful for operations that require the computer or a peripheral device for an extended period of time Real-time processing would be suitable for controlling the rockets Real-time processing causes a response within specified time constraints Real-time responses are in the order of milliseconds, and sometimes microseconds Real-time means that the inputs are processed and produce an output which in turn affects the input Controlling rockets often involves the use of sensors and control systems A computer system used for real-time processing is often used 24 hours a day for the same task Real-time data processing gives the scientists the ability to take immediate action for those times when acting within seconds is significant If a rocket is off course for just a short period of time its speed is such it would be off course by a large distance If the rocket veers off course the computer would immediately fire engines to correct it.	**8**

November 2017 – P13

13 When an expert system is used to suggest solutions to problems the inference engine uses a form of reasoning involving forward chaining or backward chaining.

Explain the differences between forward chaining and backward chaining by describing both terms.

Answer:

13	***Six*** *from:* Forward chaining starts with the available data and uses inference rules to extract more data... ... until a goal is reached An inference engine using forward chaining searches the inference rules until it finds one where the IF statement is known to be true When such a rule is found, the inference engine uses the 'THEN' part to cause the addition of new information Inference engines will iterate through this process until a goal is reached Backward chaining starts with a list of goals/hypotheses and works backwards An inference engine using backward chaining would search the inference rules until it finds one which has a THEN part that matches a desired goal If the IF part of that rule is not known to be true, then it is added to the list of goals Because the list of goals determines which rules are selected and used, this method is called goal-driven… …in contrast to data-driven forward-chaining.	**6**

March 2017 – P12

7 Atat Iron Ltd uses computers to process its payroll. The company pays its workers weekly which involves the updating of a master file.

Two sets of data are shown below.
The first set represents part of a transaction file containing workers' ID numbers and the hours worked by those workers in a particular week.
The second set represents part of the master file used by the company. This shows the workers' ID numbers, departments they work in and the rate per hour at which they are paid in Indian Rupees (₹).

Transaction file

Workers_ID_number	Hours_worked
047006	40
486439	40
592786	38
758789	40
512759	37
869891	40
471584	38
131654	40
243303	40
235804	35

Master file

Workers_ID_number	Department	Hourly_rate (₹)
031597	Extrusion	62
047006	Cold rolling	55
131654	Extrusion	62
235804	Foundry	50
239412	Foundry	50
243303	Hot rolling	58
471584	Cold rolling	55
486439	Tube production	65
500368	Extrusion	62
512759	Tube production	65
592786	Foundry	50
758789	Tube production	65
869891	Extrusion	62
942378	Hot rolling	58

(a) Describe what processes must happen before the updating can begin.

Answer:

7(a)	***Two** from:* Transaction file is validated Transaction file must be sorted… …in same order as master file/sorted on Workers ID number	**2**

Question 7, Part (b) and (c)

(b) Using this data, explain how a transaction file is used to update a master file in a payroll system. You may assume that the only transaction being carried out is the calculation of the weekly pay before tax and insurance deductions.

Answer:

7(b)	***Six** from:* First record in the transaction file read belonging to 047006 Reads first record in the old master file belonging to 031597 These two records are compared If records do not match computer writes master file record to new master file Records do not match so next record of master file is read 047006 If it matches transaction is carried out Computer calculates the pay rate of pay · no. of hours worked, 55 · 40… …using rate of pay 55 from master file… …using hours worked 40 from transaction file Processed record is written to new master file Next record 131654 is read from transaction file then compared to next master file record 131654 This continues until the last record from the transaction file record 869891 is read After processing the last record of the transaction file 869891 all the remaining old master file records are written to the new master file in this case, one record 942378	**6**

(c) When a new worker is added to the master file, the data must be validated. His hourly rate will be 50 Rupees.

Without using a type check, describe **three** other validation checks you would develop to make sure **all** the data entered is sensible.

Answer:

7(c)	Three matched pairs from:	
	Length check on Workers ID number	**1**
	Must be exactly 6 characters long	**1**
	Lookup check on Department	**1**
	Must be one of Foundry, Cold rolling, Tube production, Extrusion or Hot rolling	**1**
	Range check on hourly rate	**1**
	E.g. be between 50 and 65	**1**

June 2018 – P11

6 Many organisations use expert systems as a diagnostic tool.

Describe, in detail, the components of an expert system.

Answer:

6	**Six** *from*: Knowledge base comprises a database of facts and a rules base A database of facts is built up by the engineer on information and knowledge of the subject specialists (experts)... ...also data from databases that may exist for the topic Rules base is a set of rules which are usually of the form IF . . . THEN The shell often includes the user interface, explanation system, inference engine and knowledge base editor Inference engine is the reasoning part of the system The method used by the inference engine involves the use of forward chaining, backward chaining or a combination of both User interface is how the computer interacts with the user, displaying questions and information on a screen/enables the user to type in answers to the questions Knowledge base editing software/knowledge base editor enables the knowledge engineer to edit rules and facts within the knowledge base Explanation system explains to a user the chain of reasoning used to arrive at a particular conclusion.	**6**

June 2018 – P12

9 A medical centre uses expert systems to help diagnose illnesses.

Explain how the components of an expert system would be used to produce the diagnosis of an illness.

Answer:

9	**Six** *from*: A rules base is a set of rules which an inference engine uses The inference engine uses the data or facts in the knowledge base, to reason through the symptoms Inference engine is able to find possible diagnoses by using a form of reasoning The reasoning involves forward chaining, backward chaining or a combination of both User interface asks questions (about illness) Patient/doctor type in answers/types in symptoms to the user interface Inference engine compares symptoms to those in the knowledge base Inference engine uses the rules base of IF…THEN… rules/comparisons/ rules base consists of IF…THEN rules Knowledge base editor enables the knowledge engineer to edit rules/and facts (within the knowledge base) Description of forward chaining Description of backward chaining <u>Possible/suggested</u> diagnoses are output to user interface Explanation system produces reasons for suggestions Reasons for suggestions output using the user interface.	**6**

June 2019 – P12

9 An expert system consists of many components.

Describe how the knowledge base is used to interact with the other components when an expert system is used to solve a problem.

Answer:

9	**Five** *from:* Knowledge base consists of a database of facts / factual data and a rules base Inference engine uses the rules base to reason through the symptoms Inference engine uses the data or facts in the knowledge base to reason through the symptoms Inference engine compares symptoms to those in the knowledge base Inference engine uses the rules base of IF…THEN… rules / comparisons Knowledge base editor enables the knowledge engineer to edit rules and facts within the knowledge base.	**5**

November 2019 – P11 & P13

10 Expert systems often use a mixture of forward chaining and backward chaining to determine the probable solution to a problem.

Describe the terms:

(a) Forward chaining.

Answer:

10(a)	**Three** from: Forward chaining starts with the available data and uses inference rules to extract more data ... ... until a goal is reached An inference engine using forward chaining searches the inference rules until it finds one where the IF statement is known to be true When such a rule is found, the inference engine uses the 'Then' part to cause the addition of new information Inference engines will iterate through this process until a goal is reached Because the data entered determines which rules are selected and used, this method is called data-driven	**3**

(b) Backward chaining.

Answer:

10(b)	**Three** from: Backward chaining starts with a list of goals/hypotheses and works backwards An inference engine using backward chaining would search the inference rules until it finds one which has a THEN part that matches a desired goal If the IF part of that rule is known to be true, then it is added to the list of goals Because the list of goals determines which rules are selected and used, this method is called goal-driven	**3**

November 2019 – P12

4 Doctors often use expert systems to help them make a diagnosis.

Without describing its components explain how an expert system helps the doctor come to a decision regarding the diagnosis.

Answer:

4	**Four** from: Questions are asked by system Data/symptoms would be entered by the doctor Questions based on these would be asked by the expert system Doctor enters answers to these questions The system would reason … … using IF-THEN rules System produce probabilities of diagnoses/possible diagnoses The doctor chooses the most appropriate diagnosis	**4**

March 2019 – P12

7 Mumbai Stores is a company which has a chain of shops. The company stores the customers' reference number, and the value of the goods they have purchased, in a transaction file. Any payments the customers have made on their account at the end of the month are also stored in the same file.

Parts of the Transaction file and Master file are shown.

The Goods_bought field contains only the value of goods bought using the account's credit facilities.

The Master file shows the customers reference number, the money they owe from the previous month (Old_balance) and the money they now owe (New_balance).

The New_balance has to be calculated by using the Old_balance, Goods_bought and any Payment made. All these values are in rupees.

Transaction file

Customer_ref_no	Goods_bought	Payment
943201	5500	4000
256431	3800	3800
319852	9200	10000
433303	3200	5200
590871	4300	6800
612759	3700	3700
650897	2300	2300
698915	4000	6600
751654	4400	4400
804782	5200	8200

Master file

Customer_ref_no	Old_balance	New_balance
256431	0	
319852	10000	
433303	5000	
590871	4500	
599812	0	
612759	0	
650897	0	
678945	6800	
698915	2600	
712356	5100	
751654	0	
804782	4200	
943201	0	
963584	2400	

Question 7, Part (a) and (b)

(a) Describe the processes that must happen to the Transaction file before the Master file can be updated.

Answer:

7(a)	**Two** *from:* Transaction file is validated Transaction file must be sorted… …in same order as master file/sorted on Customer_ref_no	**2**

(b) Using this data, explain how the Master file is updated using the Transaction file shown. You may assume that the only transaction being carried out is the calculation of the New_balance.

Answer:

7(b)	**Six** *from:* First record in the transaction file read… …belonging to 256431 Reads first record in the old master file… …belonging to 256431 These two records are compared If records do not match computer writes master file record to new master file… …next record is read from master file and these two records are compared and process is repeated Records match so transaction is carried out Computer calculates the New_balance, Old_balance + Goods_bought – Payment… …0 + 3800 – 3800 (0) Using Old_balance, 0 from master file Using Goods_bought 3800, Payment made, 3800 from transaction file Processed record is written to new master file Next record is read from transaction file… …belonging to 319852 then compared to next master file record… ... belonging to 319852 This continues until the last record from the transaction file record is read … …belonging to 943201 After processing the last record of the transaction file… …belonging to 943201… …all the remaining old master file records are written to the new master file… …in this case, one record 963584.	**6**

Question 7, Part (c)

(c) Each customer has a credit limit of 20000₹. Customers must make a payment of at least 2000₹ each month.

Using examples from the Transaction file, explain why it would be appropriate to use a range check on one field and a limit check on another field.

Please note, the Customer_ref_no is stored as text.

Answer:

7(c)	**Five** *from e.g.:* Use a range check on the Payment field between 2000 and 20000 As customers must make a payment of at least 2000₹ and customers have a credit limit of 20000₹ Have a limit check on the Goods_bought field as they cannot be more than 20000₹… …customers have a credit limit of 20000₹ It is impossible to spend less than 0₹… …so a lower limit to the range is unnecessary A limit check has only one limit but a range check has two – an upper and lower limit There are alternative answers if fully reasoned.	**5**